



VLSI Testing 積體電路測試

Testability Measure

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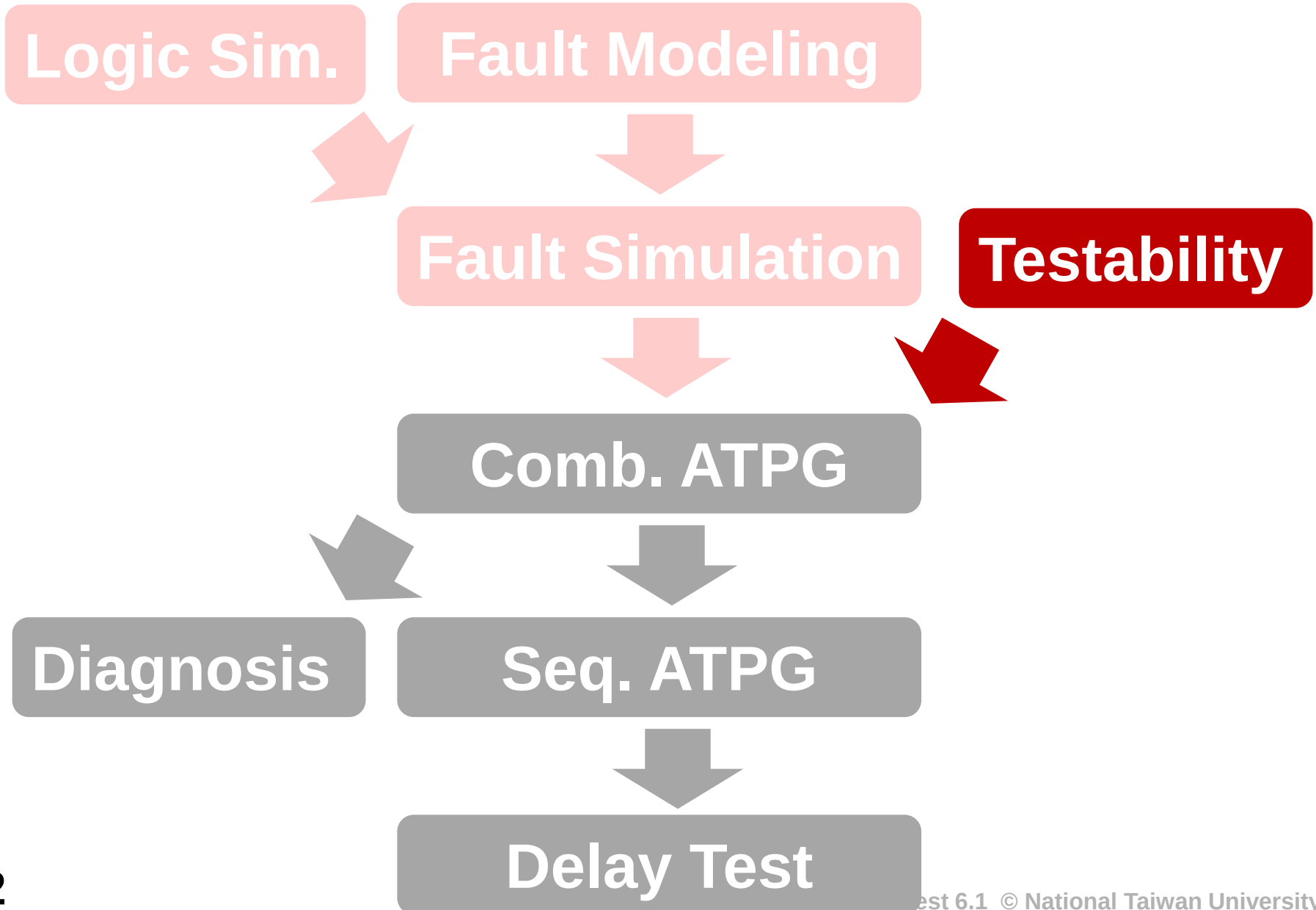
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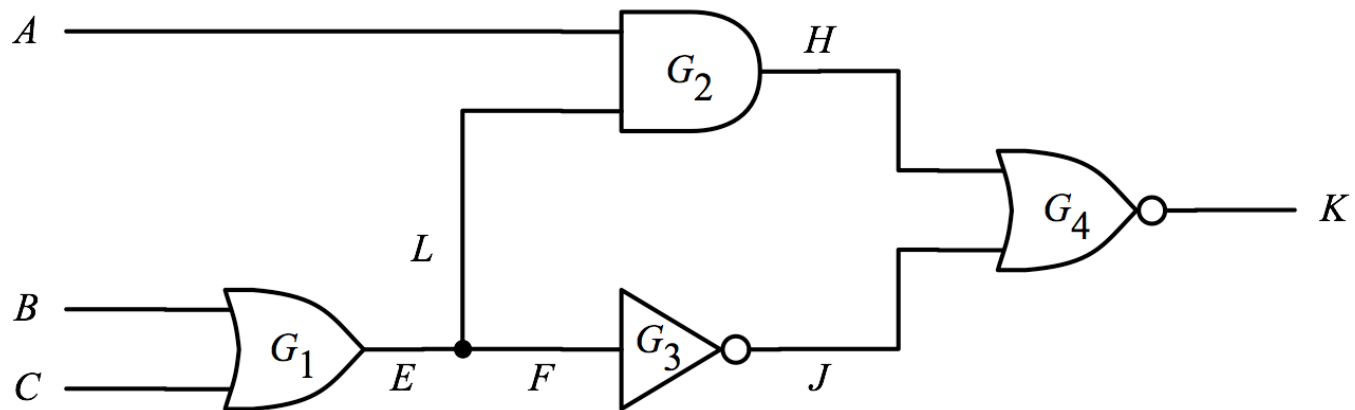
* Some pictures are courtesy of Prof. Jiun-Lang Huang, NTU

Course Roadmap (EDA Topics)



Motivating Problem

- You report fault coverage of test set to your manager. Your manager is not very happy about FC number. He asked you: which faults are so difficult to detect?



Why Am I Learning This?

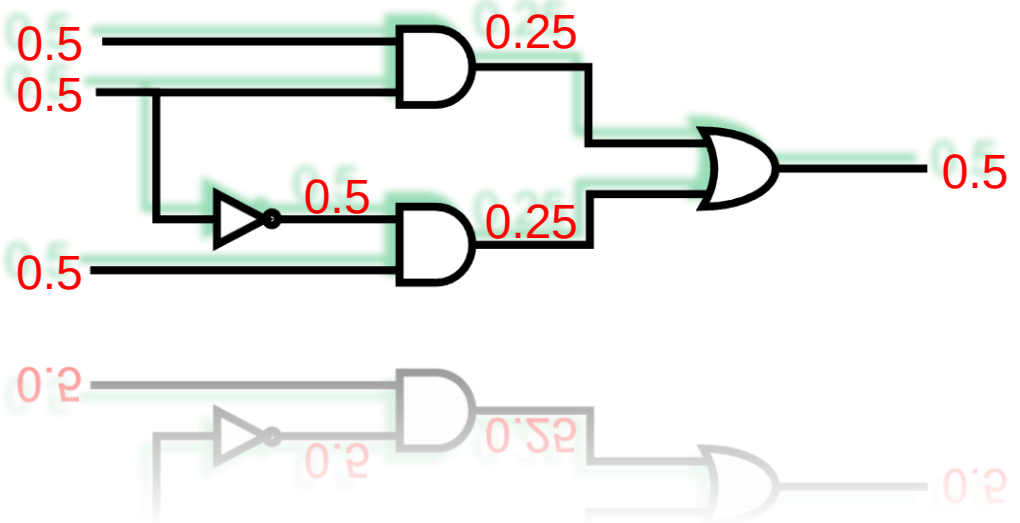
- Testability measure helps to
 - ♦ Make smart decision in ATPG
 - ♦ More testable designs

“Measurement is the first step that leads to control and eventually to improvement. ”

(H. James Harrington)

Testability Measure

- Introduction
- SCOAP (1979)
- COP (1984)
- High-level Testability Measures
- Conclusion



Testability Measures

- **What is testability measure?**
 - ♦ Metric to measure degree of difficulty to test a circuit
- **Two important components:**
 - ♦ **Controllability**
 - * degree of difficulty to control a logic signal to 0 or 1
 - ♦ **Observability**
 - * degree of difficulty to observe the logic value of a signal

Testability Measures Controllability and Observability

Testability Analysis

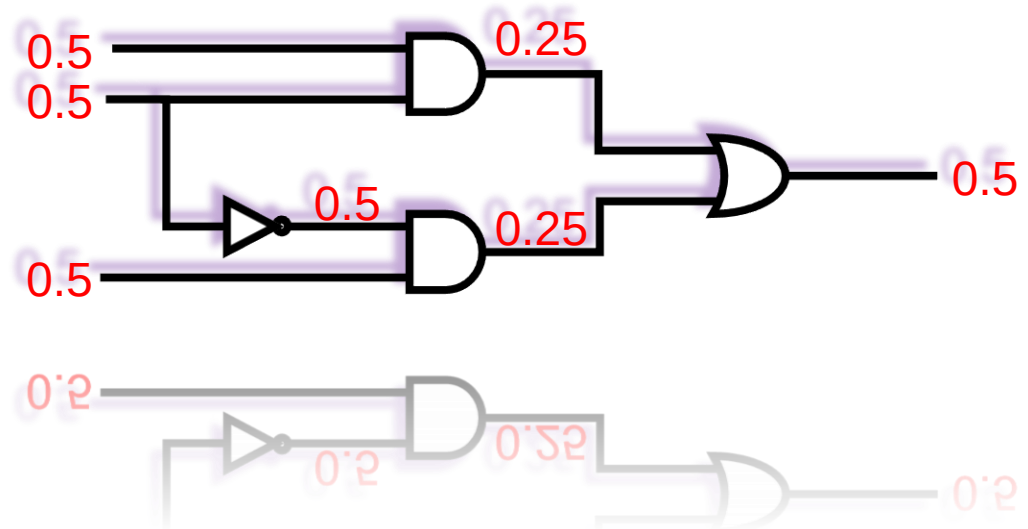
- What is *Testability Analysis*?
 - ♦ Calculate testability measures for a given circuit
- Why testability analysis?
 - ♦ 1. Help ATPG make smart decision
 - ♦ 2. Insert DFT circuits to improve controllability and observability
- When testability analysis?
 - ♦ In preprocess stage of ATPG or DFT insertion
- Requirements of testability analysis
 - ♦ Should run very fast
 - ♦ Sometimes , accuracy is not very important

Categories of Testability Analysis

- 1. **Topology-based** analysis
 - ♦ Only analyzes structure of circuit. No test vectors are given.
 - ♦ Example: SCOAP
- 2. **Probability-based** analysis
 - ♦ Uses *signal probability* to estimate the testability
 - ♦ Example: COP
- 3. **High-level** analysis
 - ♦ Performs testability analysis before synthesis
 - ♦ Example: RTL testability analysis
- 4. **Simulation-base** analysis (not in lecture)
 - ♦ Apply input patterns,
 - ♦ Perform simulation and estimate testability

Testability Measure

- Introduction
- SCOAP (1979)
 - ♦ Combinational
 - ♦ Sequential
- COP (1984)
- High-level testability measures



SCOAP [Goldstein 1979]

- **Sandia Controllability Observability Analysis Program***
- SCOAP computes 6 numbers for each node N

*Sandia is name of research lab.

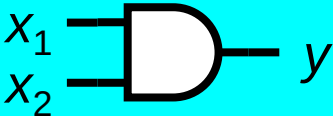
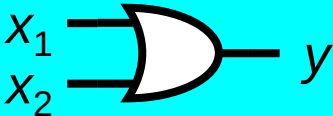
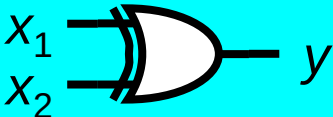
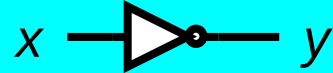
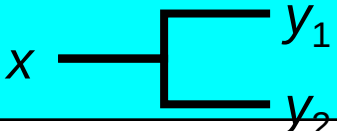
	0-controllability	1-controllability	Observability
Combinational	$CC^0(N)$	$CC^1(N)$	$CO(N)$
Sequential	$SC^0(N)$	$SC^1(N)$	$SO(N)$

Combinational Controllability

- $CC^0(N)$, $CC^1(N)$
 - ♦ minimum number of combinational PI assignments and logic levels required to control a 0 or a 1 on node N
 - ♦ *Smaller* number, *easier* to control
- How to calculate CC?
 - ♦ $CC(PI) = 1$.
 - ♦ From PI to PO. Add 1 to account for logic level
- Gate propagation rules:
 - ♦ If only one input controls gate output:
 - * $CC(\text{gate_output}) = \min \{ CC(\text{gate_input}) \} + 1$
 - ♦ If all inputs needed to set gate output:
 - * $CC(\text{gate_output}) = \sum CC(\text{gate_input}) + 1$
 - ♦ $CC(\text{branches}) = CC(\text{stem})$

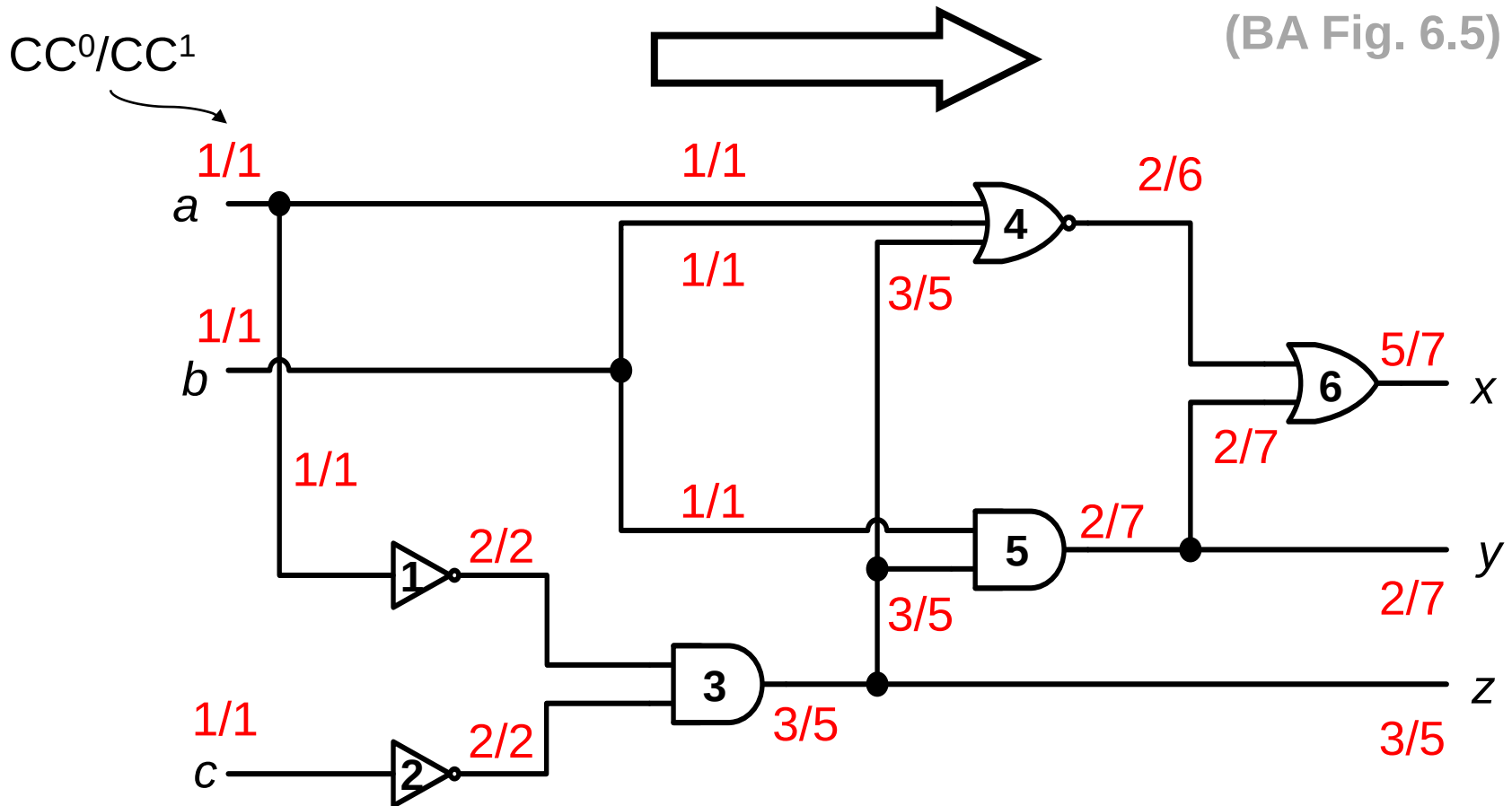
Smaller CC is Better

CC⁰(N) & CC¹(N)

	CC ⁰ (y)	CC ¹ (y)
Primary inputs	1	1
	$\min[CC^0(x_1), CC^0(x_2)] + 1$	$CC^1(x_1) + CC^1(x_2) + 1$
	$CC^0(x_1) + CC^0(x_2) + 1$	$\min[CC^1(x_1), CC^1(x_2)] + 1$
	$\min[CC^0(x_1) + CC^0(x_2), CC^1(x_1) + CC^1(x_2)] + 1$	$\min[CC^0(x_1) + CC^1(x_2), CC^1(x_1) + CC^0(x_2)] + 1$
	$CC^1(x) + 1$	$CC^0(x) + 1$
	$CC^0(y_1) = CC^0(y_2) = CC^0(x)$	$CC^1(y_1) = CC^1(y_2) = CC^1(x)$

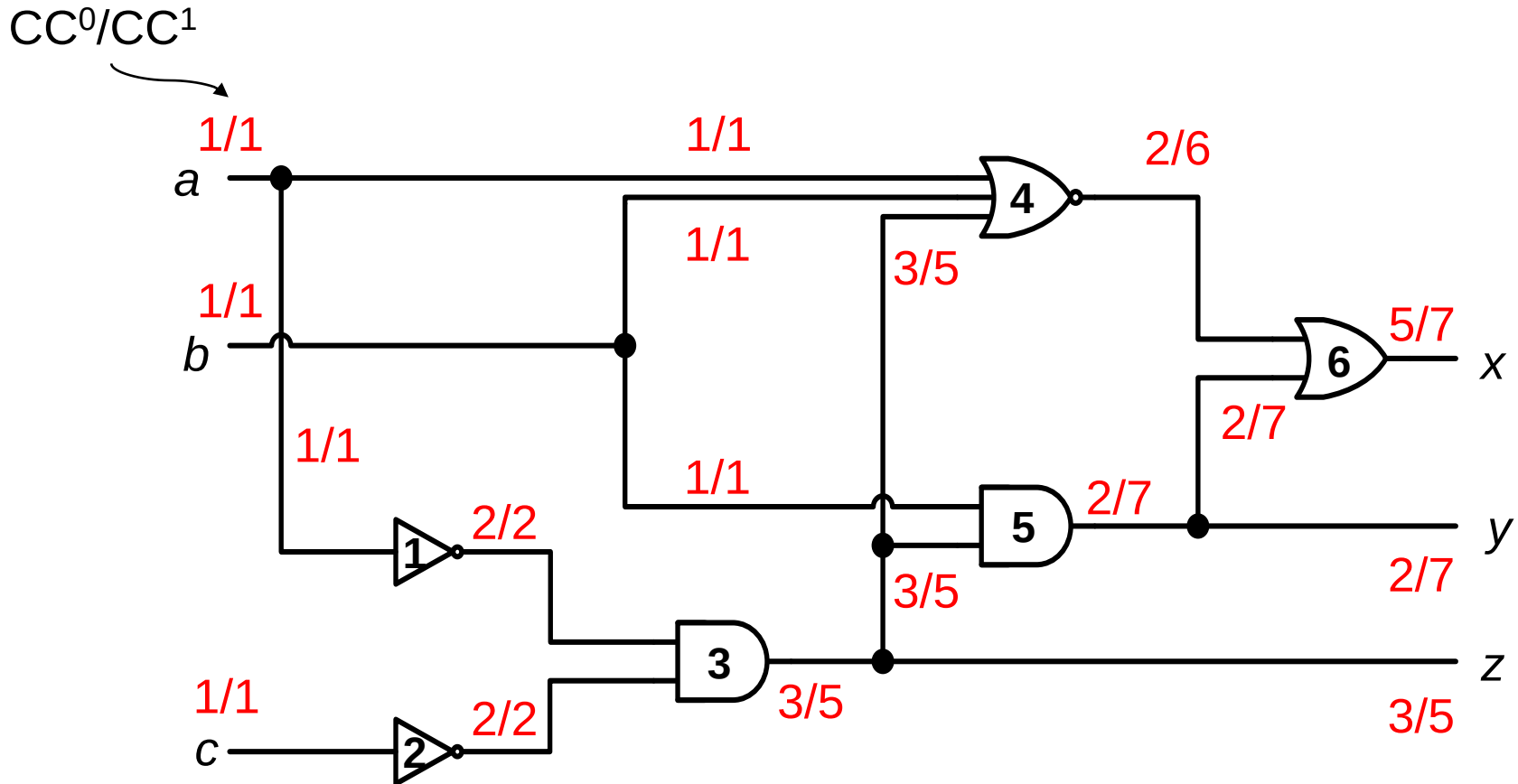
An Example – Controllability

- Forward: From PI to PO



Quiz

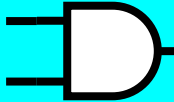
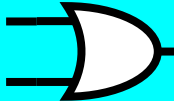
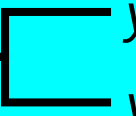
Q: how to control y to 0? How to control y to 1?



Combinational Observability

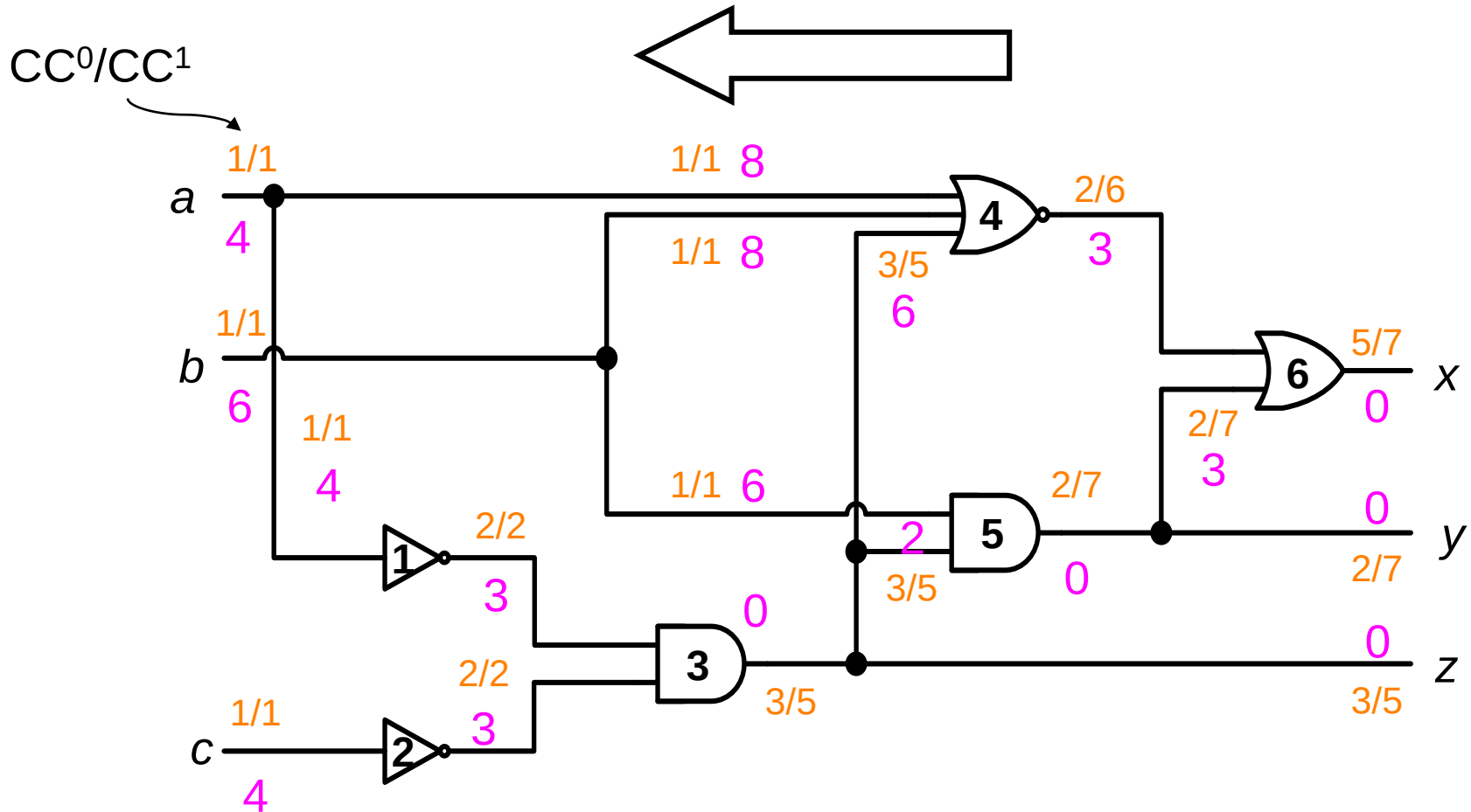
- **CO(N)**
 - ♦ minimum number of combinational PI assignments and logic levels required to propagate logical value on node N to PO
 - ♦ *Smaller* number, *easier* to observe
- **How to calculate CO(N)?**
 - ♦ $CO(PO)=0$
 - ♦ From POs to PIs, add 1 to account for logic level
- $CO(\text{gate_input}) = \Sigma$
 - ♦ (1) $CO(\text{gate_output})$
 - ♦ (2) CC(setting all other inputs to non-controlling value)
 - ♦ (3) + 1 for logic level
- **How about fanout stem?** Assume they are *independent*
 - ♦ $CO(\text{stem}) = \min \{ CO(\text{branches}) \}$ *independent*
 $CO(\text{stem}) = \max \{ CO(\text{branches}) \}$ *dependent*

CO(N)

	CO(x_1)
Primary outputs	0
x_1 x_2  y	$CO(y) + CC^1(x_2) + 1$
x_1 x_2  y	$CO(y) + CC^0(x_2) + 1$
x_1 x_2  y	$CO(y) + \min[CC^0(x_2), CC^1(x_2)] + 1$
x_1  y	$CO(y) + 1$
x_1  y_1 y_2	$\min[CO(y_1), CO(y_2)]$

An Example – Observability

- Backward: From PO to PI

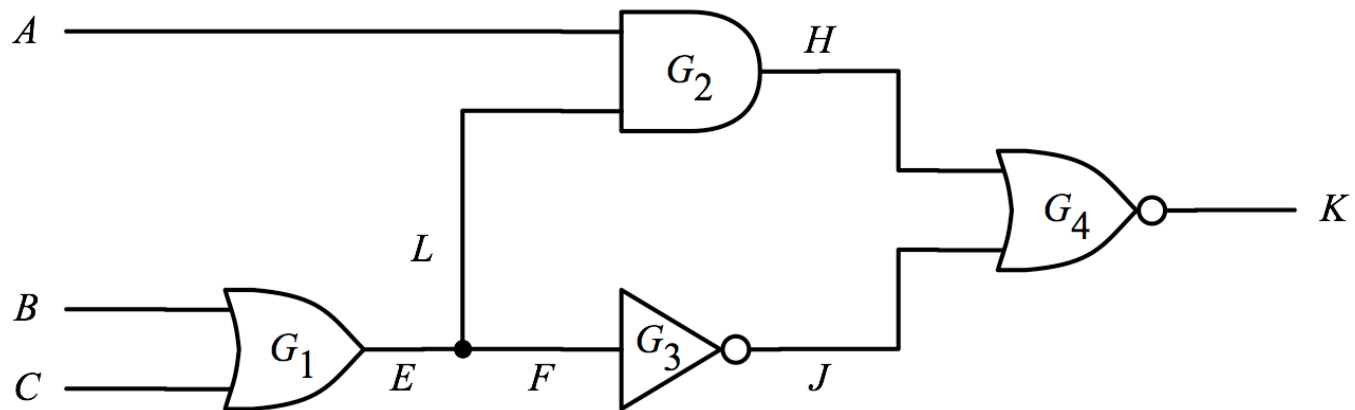


Quiz

Q: Please analyze combinational SCOAP.

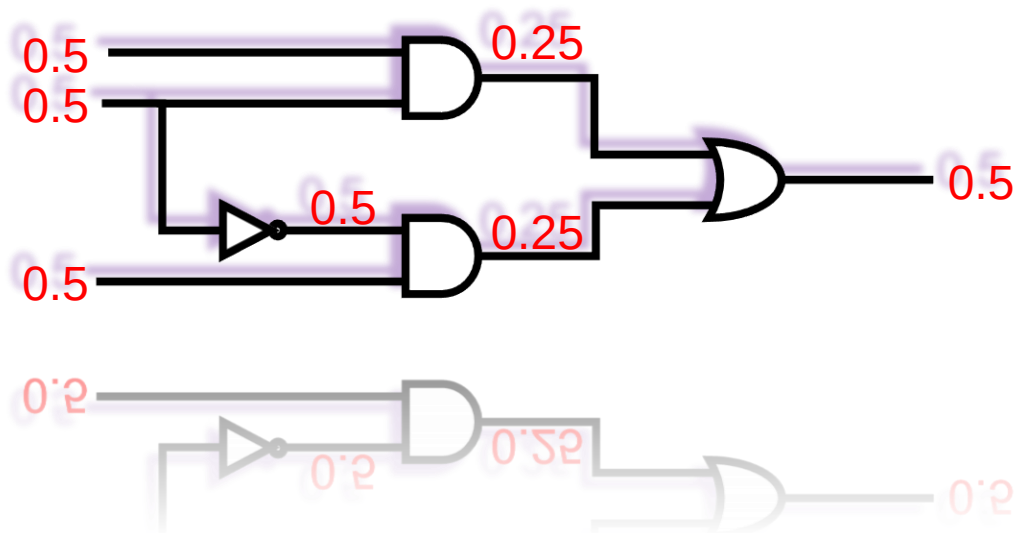
A:

	A	B	C	E	F	L	H	J	K
CC ⁰									
CC ¹									
CO									



Summary

- Introduction
 - ♦ Metric to **measure degree of difficulty to test a circuit**
 - ♦ Helps **1. ATPG 2. DfT**
- SCOAP: Combinational
 - ♦ **CC^0 , CC^1 , CO**



FFT

- Q: Testability should be done very quickly. What is time complexity to calculate CC and CO?

